Aleksandra Stamper

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EDUCATION

Brown University, *Ph.D., Epidemiology* — Providence, RI | 2023 – present
Awarded NIH Ruth L. Kirschstein National Research Service Award (F31) of $232,118 | Published 5 peer-reviewed manuscripts (3/5 first author) | Awarded NASA Rhode Island Space Grant Consortium Fellowship of $23,159 | Awarded $500 for Outstanding Doctoral Student Publication

Johns Hopkins University, *M.A.S. in Spatial Analysis for Public Health* — Baltimore, MD | 2020 – 2022
Awarded OPAL Scholarship covering $446/credit

Smith College, *B.A. in Biochemistry, Anthropology* — Northampton, MA | 2014 – 2018
Awarded $2,200 Praxis funding to support summer internship | Dean's list

TECHNICAL PROFICIENCIES

High proficiency: R (commonly used packages: ggplot2, dplyr, deSolve), SLURM, Linux/Unix, HPC environments
Familiarity: Python, SAS, SQL, ArcGIS, EPIC EMR, Tableau

PROFESSIONAL EXPERIENCE

Brown University School of Public Health — Providence, RI | 2023 – present
Graduate Research Assistant

- Publish 5 peer-reviewed manuscripts (3/5 first author) and present findings at national conferences
- Implement Susceptible-Infected-Recovered (SIR) models and fixed effects regression to investigate the impact of climate on influenza transmission and outbreak dynamics
- Use Python and R to query, process, and analyze high-dimensional climate, spatial, demographic, and infectious disease data

Georgetown University Center for Global Health Science and Security — Georgetown, DC | 2026 – present
Lead Analyst, Wastewater Epidemiology

- Analyze wastewater data testing 27 different pathogens across approximately 1,800 sewer sheds to identify emerging trends following Git practices for collaborative R coding
- Perform ad hoc analyses on geographically and temporally concentrated wastewater pathogen detections and visualize results (e.g., plots, maps) to effectively communicate findings
- Write and edit situation reports describing pathogen wastewater trends for a 1,200+-person daily distribution list (>70% open rate) and twice-daily video briefings

Brown University School of Public Health — Providence, RI | Spring 2026
Graduate Teaching Assistant

- Develop and lead five 1.5-hour coding labs in R and SAS on analyzing survey data (covering regression and assumption testing, weighting approaches to account for sampling variation, and data visualization)
- Deliver a 2.5-hour guest lecture on "Case Estimation and Global Burden"
- Grade student submissions, including providing feedback on written reports and evaluating R and SAS code for projects

University of Michigan Health Service — Ann Arbor, MI | 2021 – 2023
Health Data Analyst

- Mine, analyze, and disseminate operational, clinical, and communicable disease data as informative figures and tables to university and health system leadership
- Query, process, and visualize health record data using R, Python, SQL and tableau to support administrative and financial decision-making
- Present research findings at national conferences and publish 2 peer-reviewed manuscripts

University of Virginia Medical School — Charlottesville, VA | 2018 – 2020
Laboratory Specialist

- Investigate effects of a CRISPR-Cas9 knockout on the mouse NXF1 gene and maintain mouse colony data
- Perform molecular assays and cell culture techniques including transfection, RNA/DNA isolation, polymerase chain reaction (PCR), gel electrophoresis, Western blotting, and sequencing analysis
- Presented findings at an international conference

ORGANIZATIONS

Institute at Brown for Environment and Society, Brown University 2023 – present
Society for Epidemiologic Research 2023 - 2026

PEER-REVIEWED PUBLICATIONS

1. Edwards, E. A., McEvoy, E., **Stamper, A. R.**, Fry, J., Fleming, E., Mortenson, L. Y., & Frank, C. J. (2026). An analysis of emergency department visits for injuries among students at a large U.S. public university. *Injury Epidemiology*. https://doi.org/10.1186/s40621-026-00724-y
2. **Stamper, A. R.**, Yang, W., Wagner, C. E., Mahmud, A. S., & Baker, R. E. (2026). Unified climate factors predict influenza outbreak seasonality across tropical and temperate regions. *PNAS Nexus*, *5*(6), pgag160. https://doi.org/10.1093/pnasnexus/pgag160
3. Baker, R. E., **Stamper, A. R.**, Burrows, H. A., Spacey, O. G., Chung, M. V., Rice, B. L., Yang, W., Wagner, C. E., & Metcalf, C. J. E. (2026). Climate change and infectious diseases. *Nature Medicine*, *32*(5), 1634–1645. https://doi.org/10.1038/s41591-026-04377-8
4. **Stamper, A.,** & Baker, R. E. (2026). Spatial patterns and environmental influences of COVID-19 outbreaks, post-Omicron. *PLOS ONE*, *21*(2), e0342510. https://doi.org/10.1371/journal.pone.0342510
5. Ury, B., **Stamper, A. R.**, & Baker, R. E. (2026). Climate influences on hospitalization patterns in Mexico: Evidence from 30 million records. *Environmental Research Communications*, *8*(1), 011008. https://doi.org/10.1088/2515-7620/ae37d9
6. **Stamper, A. R.,** Mahmud, A. S., Nuzzo, J. B., & Baker, R. E. (2025). Modeling the Impact of Climate Extremes on Seasonal Influenza Outbreaks Across Tropical and Temperate Locations. *GeoHealth*, *9*(4), e2024GH001138. https://doi.org/10.1029/2024GH001138
7. Montgomery, J. P., Marquez, J. L., Nord, J., **Stamper, A. R.**, Edwards, E. A., Valentini, N., Frank, C. J., Washer, L. L., Ernst, R. D., Park, J. I., Price, D., Collins, J., Smith-Jeffcoat, S. E., Hu, F., Knox, C. L., Khan, R., Lu, X., Kirking, H. L., & Hsu, C. H. (2024). Detection of a Human Adenovirus Outbreak, Including Some Critical Infections, Using Multi-Pathogen Testing at a Large University, September 2022—January 2023. *Open Forum Infectious Diseases*, ofae192. https://doi.org/10.1093/ofid/ofae192
8. Lewis, N. M., Delahoy, M. J., Sumner, K. M., Lauring, A. S., Bendall, E. E., Mortenson, L., Edwards, E., **Stamper, A.**, Flannery, B., & Martin, E. T. (2023). Risk factors for infection with influenza A(H3N2) virus on a US university campus, October-November 2021. *Influenza and Other Respiratory Viruses*, *17*(5), e13151. https://doi.org/10.1111/irv.13151
9. Sood, A., Miller, A. M., Brogi, E., Sui, Y., Armenia, J., McDonough, E., Santamaria-Pang, A., Carlin, S., **Stamper, A.**, Campos, C., Pang, Z., Li, Q., Port, E., Graeber, T. G., Schultz, N., Ginty, F., Larson, S. M., & Mellinghoff, I. K. (n.d.). Multiplexed immunofluorescence delineates proteomic cancer cell states associated with metabolism. *JCI Insight*, *1*(6), e87030. https://doi.org/10.1172/jci.insight.87030

ORAL PRESENTATIONS

1. **Stamper A**, Modeling the Impact of Temperature and Weather Extremes on Influenza Epidemics. Examining the Intersection of Infectious Disease and Extreme Weather. Virtual; March 2026
2. **Stamper A**, Climate and Influenza: Evaluating Drivers Across Temperate and Tropical Regions. Rhode Island Space Grant Spring Symposium. Providence, RI; April 2025
3. Mortenson L, **Stamper A**, Edwards E, Montalva R. Clinical Informatics, Metrics and KPIs: Essentials for College Health Professionals Responding to Public Health Threats. American College Health Association Meeting. Boston, MA; June 2023

CONFERENCE POSTERS

1. **Stamper A**, Baker E. Spatial patterns and environmental influences of COVID-19 outbreaks in the United States, post-Omicron. Ecology and Evolution of Infectious Diseases Conference. Blacksburg, VA; June 2026.

2. **Stamper A**, Baker E. Climate and Influenza: Evaluating Drivers Across Temperate and Tropical Climates. Ecology and Evolution of Infectious Diseases Conference. South Bend, IN; June 2025
3. **Stamper A**, Baker E. Evaluating climate drivers of influenza dynamics across tropical and temperate regions. Society for Epidemiologic Research Meeting. Boston, MA; June 2025
4. **Stamper A**, Baker E. Impact of climate variability on projected influenza outbreaks. Society for Epidemiologic Research Meeting. Austin, TX; June 2024
5. **Stamper A**, Edwards E. Why do College Students Seek Emergency Services? American College Health Association Meeting. Boston, MA; June 2023
6. Edwards E, **Stamper A**. What Happens to College Students who Seek Emergency Medical Services? American College Health Association Meeting. Boston, MA; June 2023
7. Rutkowska M, **Stamper A**, Michalak M, Xu W, Huang J, Rekosh D, Hammarskjold ML. Development and Analysis of Mouse Models to Study the Function of the Constitutive Transport Element (CTE) in the NXF1 Gene. RNA Society Meeting. Krakow, Poland; June 2019
8. Rutkowska M, **Stamper A**, Michalak M, Xu W, Rekosh D, Hammarskjold ML. A Deletion in the Mouse Nxf1 Intron 10 CTE Results in Downregulation of Genes Involved in Learning and Synaptic Plasticity. RNA Society Meeting. Krakow, Poland; June 2019

JOURNAL REVIEW

Nature Reviews
Nature Communications
PLOS Computational Biology

SERVICE

Student Director – Department of Epidemiology Doctoral Seminar, Brown University	Fall 2025, Spring 2026
Member – School of Public Health Doctoral Studies Committee, Brown University	Fall 2025, Spring 2026

TEACHING

Teaching Assistant, Surveillance Methods for Applied Epidemiology, Brown University	Spring 2026